MA 252 | FINAL COURSE PROJECT

Analysis of College Major Salaries

What predicts earnings after graduation?

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Research Question & Dataset

RESEARCH QUESTION

Which factors best explain variation in **median salary** across 172 college majors?

Specifically:

- Does gender composition (ShareWomen) predict earnings?
- Does unemployment rate matter?
- How much does field of study (Major Category) drive salary differences?

THE DATASET

172 academic majors

Key Variables:

- Median Salary (response)
- ShareWomen (proportion female)
- Unemployment Rate
- Major Category (16 categories)

n = 172 complete observations after removing NAs

Methods Overview

01

Simple Linear Regression

Baseline model: Median salary predicted by ShareWomen alone. Establishes the bivariate relationship before adding complexity.

02

Multiple Linear Regression

Expanded model adding Unemployment Rate and Major Category (as factor). Allows us to control for field of study and other variables.

03

Stepwise Model Selection

Bidirectional stepwise procedure (AIC-based) to identify the best combination of predictors from all available variables.

04

Model Assumptions

Residuals vs. Fitted, Q-Q plot, Scale-Location, Cook's Distance, and VIF analysis to validate assumptions and check for multicollinearity.

Exploratory Data Analysis

Mean Salary

$40,077

Median: $36,000

Avg. Share Women

52%

Range: 0% – 97%

Avg. Unemployment

7%

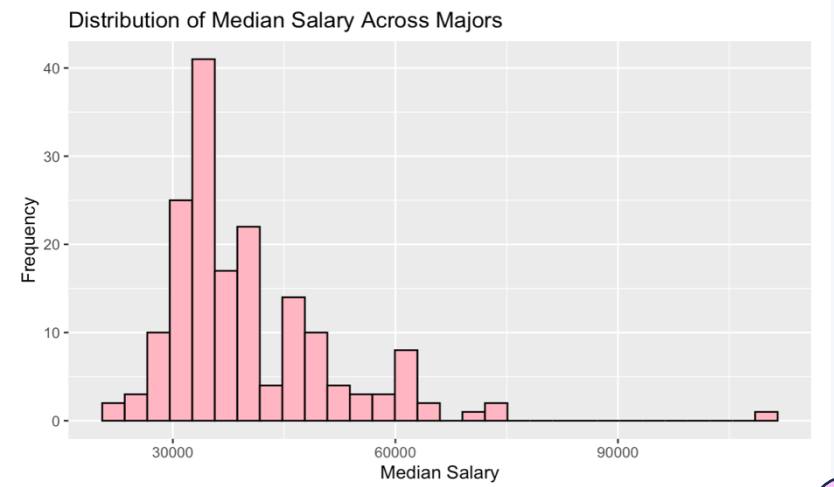
Range: 0% – 18%

Distribution of Median Salary

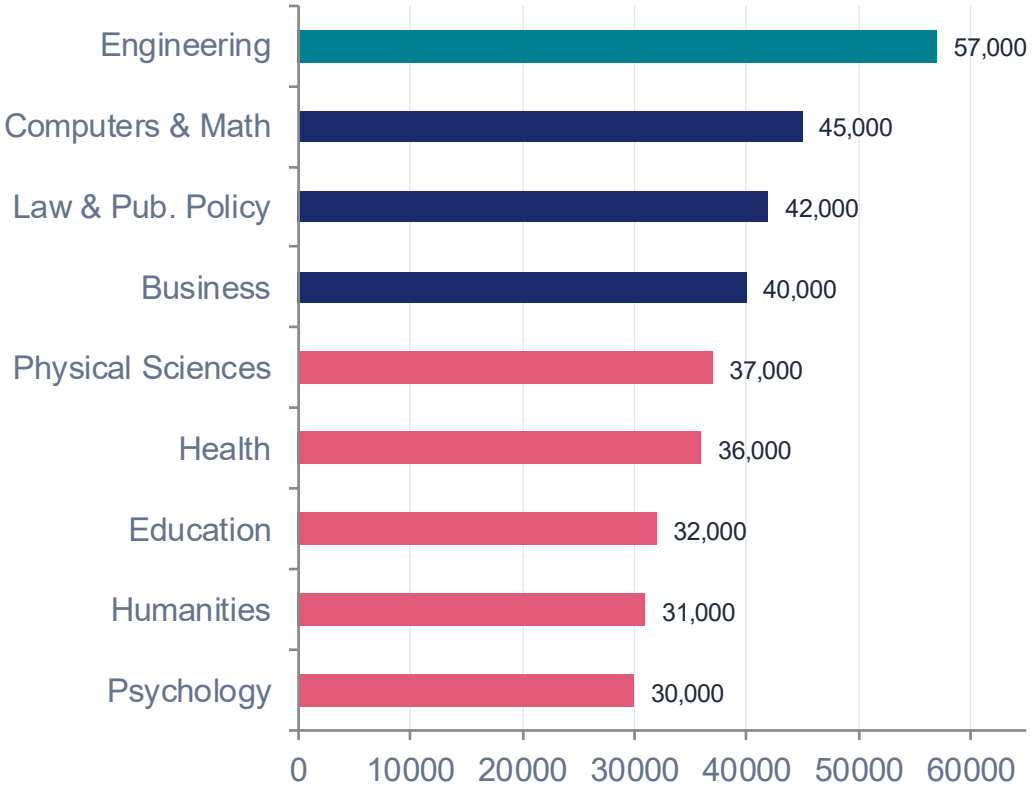
Right-skewed distribution — most majors fall between $30K–$50K, but a few high-earning fields pull the mean above the median.

The interquartile range spans $33K–$45K, meaning half of all majors fall within a ~$12K salary band.

Salary range: $22,000 (minimum) to $110,000 (maximum) — a nearly 5× spread across majors.



Salary Differs Dramatically by Field



HIGH EARNERS

Engineering majors earn ~$57K median — nearly double Education fields

LOWER EARNERS

Psychology, Education & Humanities cluster below $33K median

KEY TAKEAWAY

Major category is the number one predictor of salary — confirmed by all three models

Regression Models: Equations

MODEL 1 — Simple Linear Regression

$$\text{Median Salary} = \beta_0 - \beta_1(\text{ShareWomen})$$

$$\text{Median Salary} = 56{,}093 - 30{,}670 \times \text{ShareWomen}$$

$R^2 = 0.383$ | $p < 2e\text{-}16$ | ShareWomen alone explains 38% of salary variation

MODEL 2 — Multiple Linear Regression (Stepwise Final Model)

$$\text{Median Salary} = \beta_0 + \beta_1(\text{ShareWomen}) + \beta_2(\text{UnemploymentRate}) + \Sigma \beta_{3k}(\text{MajorCategory}_k)$$

where $\text{MajorCategory}_{ik} = 1$ if major i is in category k, 0 otherwise

$R^2 = 0.622$ | Adjusted $R^2 = 0.580$ | $F(17, 154) = 14.9$, $p < 2.2e\text{-}16$

Model 1: Simple Linear Regression and Interpretations

Median Salary = 56,093 – 30,670 × ShareWomen

β_0

$56,093

Predicted salary for a major with 0% women

β_1

–$30,670

Each 1-unit increase in ShareWomen → salary drops by $30,670

R^2

0.383

ShareWomen alone explains 38% of salary variation

p-value

2e-16 < α

Relationship is highly statistically significant since less than 0.05 (alpha)

Model 2: Multiple Linear Regression

$$\text{Median Salary} = \beta_0 + \beta_1(\text{ShareWomen}) + \beta_2(\text{UnemploymentRate}) + \Sigma\, \beta_{3k}(\text{MajorCategory}_k)$$

ShareWomen	Unemployment Rate	Major Category
−$15,238	**−$30,091**	**★ Strongest**
• Significant (p = 0.0004)	• NOT significant (p = 0.16)	• Engineering: +$20,083 (p < 0.001)
• Effect shrinks from −$30,670 in simple model	• Does not meaningfully predict salary	• Business: +$10,194 (p = 0.002)
• Gender gap persists even after controlling for field	• Once major category is controlled for	• Computers & Math: +$7,186 (p = 0.04)

R^2 improved from 0.383 (simple model) → 0.622 (full model) | Model now explains 62% of salary variation

E

Stepwise Model Selection & VIF

How the stepwise algorithm (bidirectional, AIC-based) built the final model:

Step 1

Major Category

AIC: 3096

Biggest AIC drop — strongest predictor

Step 2

+ ShareWomen

AIC: 3083

Meaningful gain — included

Step 3

+ Unemployment Rate

AIC: 3083

Marginal gain — included

Excluded

✕ Total Graduates

No AIC gain

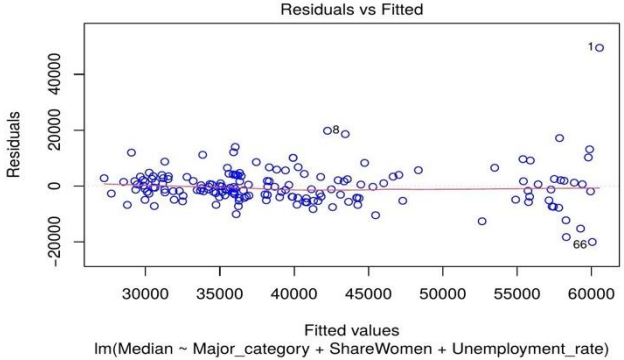
Not selected — no improvement

Multicollinearity Check (VIF) — All values well below cutoff of 5

Predictor	GVIF	Adjusted VIF	Concern?
Major Category	3.75	1.045	✓ None
ShareWomen	2.93	1.713	✓ None
Unemployment Rate	1.30	1.139	✓ None

Model Diagnostics: Residual Plots

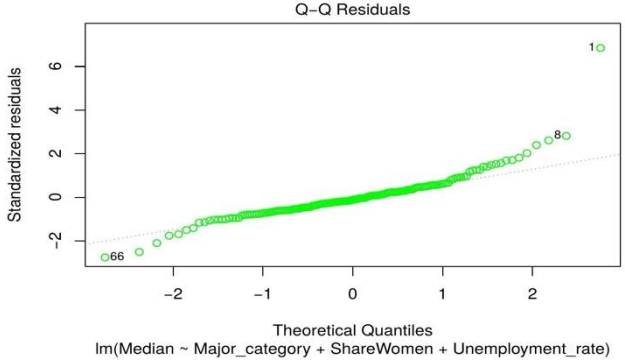
Residuals vs. Fitted



✓ Linearity met

Residuals scatter evenly around zero with no strong curvature. Slight variance increase at higher fitted values — mild heteroscedasticity, not severe.

Normal Q-Q Plot



✓ Normality approximately met

Most points follow the reference line. Some deviation in the upper tail (obs. 1, 8) from high-earning outliers, but normality assumption holds overall.

Model Comparison: Is the Full Model Better?

Nested ANOVA: $F(16, 154) = 6.09$, $p = 2.82 \times 10^{-10}$ → Full model is significantly better ✓

WHAT THIS MEANS

62% explained vs 38% in the simple model

Adding Major Category and Unemployment Rate captured an extra 24% of salary variation.

The ANOVA confirms this improvement is statistically significant — not due to chance.

Conclusion

What We Found

• Major category is the strongest driver of salary differences across fields — Engineering, CS, and Business earn significantly more.

• Gender composition (ShareWomen) remains a significant negative predictor ($p = 0.0004$) even after controlling for field — suggesting a within-field gender earnings gap.

• Unemployment rate adds little explanatory power once major category is accounted for ($p = 0.16$).

• Our final model explains 62% of salary variation — a major improvement over 38% with gender alone.

Why It Matters

Choosing a major has lasting financial consequences. Understanding which factors drive salary — and which don't — helps students and advisors make better decisions.

Future Directions

• Explore interaction terms (e.g., gender × field)

• Include region or cost-of-living adjustments

• Examine graduate-level vs. recent-grad data

Key Takeaways

01

Major Category is the #1 Predictor

Field of study dominates salary outcomes. Engineering, Business, and CS majors earn significantly more — effect is large and highly significant.

02

Gender Gap Persists Across Fields

ShareWomen stays significant ($p = 0.0004$) even after controlling for major — suggesting a within-field gender earnings gap.

03

Unemployment Rate Adds Little

Not significant once major category is included ($p = 0.16$) — field of study absorbs most of its explanatory power.

04

Full Model Explains 62% of Variation

Major improvement over the simple model ($R^2 = 0.383 \rightarrow 0.622$), confirmed by nested ANOVA ($F = 6.09$, $p < 0.001$).

References

DATA SOURCE

FiveThirtyEight. (2014). Recent College Graduates Dataset [recent-grads.csv].

Retrieved from: https://github.com/fivethirtyeight/data/tree/master/college-majors

R PACKAGES

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Fox, J. & Weisberg, S. (2019). car: Companion to Applied Regression. R package version 3.1-2.

Robinson, D. & Hayes, A. (2023). moderndive: Statistical inference via data science. R package version 0.5.4.

Waring, E. et al. (2022). skimr: Compact and Flexible Summaries of Data. R package version 2.1.5.

COURSE MATERIAL

MA 252 course notes and lectures — Normal Regression, model selection, and diagnostics.